

Curated Research Articles

Generated: 2026-09-28 08:34

- **Materials-Property Classification of Interlayers for Stabilizing Lithium Metal Interfaces in Sulfide-Based All-Solid-State Batteries** — score: 1.000 The article presents a property-guided framework for designing interlayers to stabilize lithium metal interfaces in sulfide-based all-solid-state batteries (ASSLMBs), which face interfacial instability issues. By categorizing interlayer materials—such as metals, polymers, ceramics, and composites—based on their mechanical and electrochemical properties, the study emphasizes how these characteristics influence interfacial behavior and degradation, ultimately aiding the rational design of stable battery interfaces. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Multi-modal Operando Analysis of Lithium–Sulfur Batteries: A Comparative Study on the Influence of Electrolyte Composition** — score: 1.000 The article presents a comparative study utilizing a multimodal operando analysis to explore how different electrolyte compositions affect the performance and degradation mechanisms of lithium-sulfur batteries. Through techniques such as Raman and UV–vis spectroscopy alongside impedance measurements over 20 cycles, the research reveals that while less effective solvents can enhance battery longevity, they simultaneously limit the availability of active polysulfides, impacting overall capacity. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Mechanistic Origin of Capacity Limitation in Sidorenkite-Type $\text{Na}_{3-x}\text{Fe}(\text{PO}_4)(\text{CO}_3)$ Carbonophosphate Cathodes for Na-Ion Batteries** — score: 1.000 The study investigates the limited capacity of sidorenkite-type $\text{Na}_{3-x}\text{Fe}(\text{PO}_4)(\text{CO}_3)$ cathodes for sodium-ion batteries, attributing this limitation to lattice strain caused by the formation of Fe^{4+} ions, which impair electronic conductivity. It proposes design strategies, such as preserving crystallinity and avoiding unstable d4 configurations, to enhance the performance of polyanionic cathodes and achieve higher energy storage. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Activating the Anomalous Jahn–Teller Effect in NASICON-Type Mixed Phosphate Cathode via Crystal Field Engineering** — score: 1.000 The study explores a novel NASICON-type cathode material, $\text{Na}_{3.85}\text{O}_{0.15}\text{Mn}_{1.35}\text{Fe}_{1.5}\text{V}_{0.15}(\text{PO}_4)_2(\text{P}_2\text{O}_7)$, which utilizes vanadium substitution to effectively manage the Jahn–Teller distortion associated with Mn^{3+} ions. This engineering leads to enhanced energy density, improved reaction kinetics, and a significant increase in sodium mobility, resulting in a high reversible capacity and strong cycle stability, thereby providing a promising approach for advancing battery technology. Journal: *Journal of the American Chemical Society ASAP*
- **Intrinsic Planar Defects Govern Durability in Single-Crystal Ni-Rich Cathodes** — score: 1.000 The study investigates the electrochemical stability of single-crystal Ni-rich cathodes in lithium-ion batteries, revealing that intrinsic planar defects, specifically disordered layered phase boundaries, lead to performance issues due to increased electrochemical variability. The authors propose that a multistep lithiation process can help heal these defects, ultimately enhancing the cathode’s efficiency and capacity retention over time. Journal: *Journal of the American Chemical Society ASAP*
- **Disentangling Lithium Failure Mechanisms in Liquid Electrolytes by 2D Exchange Solid-State NMR** — score: 1.000 This article introduces a new analytical method using two-dimensional exchange solid-state NMR (2D EXSY ssNMR) to investigate lithium failure mechanisms in liquid electrolytes, particularly focusing on lithium metal batteries. The study reveals that the failure modes are primarily linked to either electronic isolation or ionic contact loss, depending on the electrolyte composition, and highlights the significance of solid electrolyte interphase (SEI) characteristics in dictating battery longevity and efficiency. The findings emphasize the need for SEI designs that are both chemically stable and mechanically adaptable to mitigate these failure processes. Journal: *Journal of the American Chemical Society ASAP*
- **Elucidating Sodium-Ion Storage Mechanisms in Bowl-Shaped Hierarchical Porous Carbon Anodes: A Combined Solid-State NMR and EPR Approach** — score: 1.000 This study investigates sodium-ion storage mechanisms in bowl-shaped hierarchical porous carbon anodes, utilizing

solid-state NMR and EPR to reveal four distinct Na⁺ ion environments and their interactions as a function of electrochemical potential. The findings correlate the unique carbon architecture with sodium storage pathways, such as adsorption, insertion, and intercalation, thereby informing the design of enhanced anodes for sodium-ion batteries. Journal: *Journal of the American Chemical Society ASAP*

- **Tuning H₂O-Engaged Redox Reaction Pathway via Water Orientation in Aqueous Batteries** — score: 1.000 The article highlights the critical role of interfacial water orientation in enhancing the performance of aqueous batteries through a novel H₂O-engaged redox reaction pathway. By utilizing N vacancies in WN artificial interfaces to reorient water, researchers achieved significant improvements in the oxidation reaction of Mn²⁺, leading to enhanced battery capacity, ultrahigh rate capability, and extended cycling lifespan. This study represents a key advancement in the mechanistic understanding and design of aqueous battery technologies. Journal: *Journal of the American Chemical Society ASAP*
- **Online Quantification of Lattice Sodium for O3-Type Cathode Diagnostics during Scalable Manufacturing** — score: 1.000 The article introduces a novel method for the online measurement of lattice sodium in O3-type cathodes, crucial for optimizing sodium-ion battery performance. By employing the diffraction peak intensity ratio (I_{Na}/I_{Fe}), the study establishes a reliable, nondestructive way to quantify lattice sodium and predict electrochemical properties, supporting scalable manufacturing through an automated testing platform. This advancement aims to enhance the efficiency and sustainability of sodium-ion battery production. Journal: *Journal of the American Chemical Society ASAP*
- **Quantifying the Interfacial Energy Gap for Reversible Lithium–Metal Anodes** — score: 1.000 The study addresses the challenges of low Coulombic efficiency in lithium–metal batteries by proposing the interfacial energy gap as a metric for understanding interfacial electron transfer. It identifies the solvent’s minimum electrostatic potential as a key factor influencing this gap, leading to the design of new solvents that achieve over 99% efficiency. The findings offer a systematic approach for optimizing electrolyte formulations to improve the performance of lithium–metal anodes. Journal: *Journal of the American Chemical Society ASAP*
- **Deciphering Electrocatalytic Sulfur Reduction Reactions in Al–S Batteries via Single-Particle Electrodes** — score: 1.000 This article discusses the development of a single-particle electrochemical methodology to analyze the sluggish sulfur reduction reaction (SRR) in aluminum-sulfur batteries. Utilizing a cobalt/nickel and nitrogen codoped graphene electrocatalyst significantly improves SRR kinetics, particularly the conversion of Al₂S₄, which is crucial for battery performance. The findings suggest that optimizing electrocatalysis can enhance the overall energy output of these batteries. Journal: *Journal of the American Chemical Society ASAP*
- **Stabilizing High-Voltage Operation in O3-Type Na-Layered Cathodes by Suppressing Distortion-Driven Fe Migration** — score: 1.000 The article discusses a method to enhance the performance of Fe-based O3-type Na-layered cathodes by addressing the issue of voltage decay and capacity fading, which is caused by Fe migration and Jahn-Teller distortion during high-voltage operation. By optimizing the Fe content and reinforcing the transition-metal framework, the approach successfully stabilizes the cathode structure, achieving a capacity of 177.96 mAh g^{−1} and maintaining over 73% capacity retention after 100 cycles, thus providing insights into improving high-voltage applications for these materials. Journal: *ACS Energy Letters ASAP*
- **Nonmonotonic Li-Content Evolution and Size-Dependent Lithium Accessibility in Degraded High-Nickel Cathodes** — score: 1.000 The study investigates how degradation affects lithium insertion and extraction pathways in high-nickel cathodes, revealing that the evolution of lithium accessibility is nonmonotonic and varies with particle size. By employing advanced synchrotron techniques, the researchers found that larger particles face intra-particle lithium state gradients, while smaller particles show significant inter-particle differences, contributing to a complex discharge behavior that hinders battery performance. Journal: *ACS Energy Letters ASAP*
- **Mapping Solid Electrolyte Interphase Evolution and Lithium Deposition at Solid-State**

Interfaces by Operando Nano-focus WAXS — score: 1.000 The study investigates the use of anion-trapping layered double hydroxide (LDH) in composite solid polymer electrolytes (CSPEs) to enhance lithium deposition and stabilize the solid electrolyte interphase (SEI) in lithium metal batteries. Using operando nano-focus wide-angle X-ray scattering, the researchers visualized the spatial evolution of the SEI and lithium dendrites, demonstrating that the LDH-modified electrolyte leads to improved cycling stability and mitigates dendrite growth, crucial for the development of dendrite-free all-solid-state batteries. Journal: *ACS Energy Letters* ASAP

- **Enhancing the Interphasic Stability of Sodium-Ion Batteries via Structural Distortion-Driven Defluorination of Aromatic Additives** — score: 1.000 The study identifies a novel method to improve the interphasic stability of sodium-ion batteries by utilizing fluorinated aromatic additives, specifically 1,2-difluorobenzene (2FB). Upon reduction, 2FB undergoes structural changes that weaken C–F bonds, leading to defluorination and forming a protective sodium fluoride-rich solid electrolyte interphase. This layer not only enhances cycling stability and Na⁺ transport in hard carbon anodes but also mitigates issues like interphasic dissolution and sodium plating during fast charging. Journal: *ACS Energy Letters* ASAP
- **Formation and Electrochemical Role of an Ultrathin Compositionally Complex Interfacial Layer on O3-Type Layered Cathodes** — score: 1.000 The article discusses the development of a thin, compositionally complex interfacial layer made from a Na–Ni–Fe–Mn–Ti–O multicomponent oxide that improves the performance of sodium-ion batteries. This layer, created through atomic layer deposition and thermal annealing, enhances electrochemical stability and capacity retention of layered oxide cathodes by suppressing surface degradation and facilitating better Na⁺ transport, leading to significantly improved cycling performance. Journal: *ACS Applied Energy Materials* ASAP
- **Insight into Zn 2+ Storage in Ti 3 C 2 T x MXene within the Underpotential Deposition Region** — score: 1.000 This study investigates the mechanisms of Zn²⁺ storage in Ti 3 C 2 T x MXene within the underpotential deposition (UPD) region, utilizing various analytical methods. It identifies a surface-confined process that occurs before bulk Zn deposition, highlighting the adsorption and formation of a quasi-two-dimensional Zn phase associated with Ti=O terminations. The findings enhance the understanding of Zn storage dynamics and suggest pathways for improving Zn-based energy-storage technologies. Journal: *ACS Applied Energy Materials* ASAP
- **Structural Evolution of Biphenyl Phenolic Novolac Resin-Derived Hard Carbon for Enhanced Sodium Storage** — score: 1.000 The article explores the use of biphenyl phenolic novolac resin to create hard carbon anodes for sodium-ion batteries, highlighting how pyrolysis temperature influences structural properties like defect density and nanopore size. The optimally processed hard carbon exhibits impressive performance, achieving a reversible capacity of 387.7 mAh g⁻¹ and high cycling stability, while also clarifying the sodium storage mechanism through a combination of adsorption, intercalation, and filling processes. Journal: *ACS Applied Energy Materials* ASAP
- **P2-Type Cathodes Enabled by Oxygen Vacancies and Lattice Stabilization for High-Rate and Low-Temperature Performance** — score: 1.000 The article presents a novel strategy for developing P2-type cathodes in sodium-ion batteries by incorporating Mg/Ti co-doping and creating a Na-rich structure, which enhances both ionic conductivity and structural stability. This approach effectively manages oxygen vacancies, resulting in improved electrochemical performance, including significant capacity retention and superior low-temperature operation. Journal: *ACS Applied Energy Materials* ASAP
- **Solvation Chemistry and Interfacial Modulation by Zwitterionic 2-Cyclohexylaminoethanesulfonic Acid for Highly Reversible Aqueous Zinc Batteries** — score: 1.000 The study introduces 2-cyclohexylaminoethanesulfonic acid (CHES) as an innovative amphoteric additive that significantly improves the performance of aqueous zinc-ion batteries by regulating solvation chemistry and interfacial dynamics. CHES effectively suppresses dendrite growth and enhances cycling stability by forming a protective layer on the Zn anode, allowing for over 3000 hours of stable cycling and improved capacity retention in full cells. Journal: *Wiley: Small Methods: Table of Contents*

- **Ta 5d-Orbital-Mediated Local Electronic-Structure Regulation Enables Durable 4.7 V LiCoO₂ Cathodes** — score: 1.000 This study demonstrates that tantalum (Ta) enhances the stability and performance of LiCoO₂ (LCO) cathodes at high voltages (up to 4.7 V) by regulating the local electronic structure through 5d orbital interactions. The unique Ta O Co coupling reduces the overlap between Co 3d and O 2p states, preventing structural degradation and oxygen release, leading to improved cycling stability and significant retention of capacity after extensive use. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Pre-Sodiation Strategies for Sodium-Ion Batteries: From Mechanisms to a Multidimensional Evaluation Framework** — score: 1.000 The article reviews the challenges of low initial efficiency in sodium-ion batteries due to sodium ion loss, advocating for pre-sodiation as a crucial strategy to mitigate this issue. It proposes a comprehensive framework that categorizes various pre-sodiation methods—physical, electrochemical, chemical, and additive-based—into a multidimensional evaluation system to systematically assess their effectiveness. This framework considers not just electrochemical metrics but also interfacial and structural properties, providing guidelines for selecting optimal pre-sodiation approaches to enhance battery performance. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **“Spike-Accessed” Polymer-Based Interphase-Engineered Surface-to-Bulk Lattice Stabilization for Ni-Rich Positive Electrodes** — score: 1.000 The article presents a novel spike-accessed polysiloxane-based interphase designed to enhance the stability of Ni-rich layered oxide positive electrodes in lithium-ion batteries. By facilitating dynamic redox buffering and improving the coordination of nickel species, this engineered interphase significantly mitigates structural degradation during high-voltage cycling, allowing for extended battery life and performance even after thousands of charge cycles. This innovative approach not only preserves electrode integrity but also sets a new standard for interfacial engineering in energy storage systems. Journal: *Journal of the American Chemical Society ASAP*
- **Lattice Distortion Regulates Na-Ion Percolation in High-Entropy Cation-Disordered Rocksalt Cathodes** — score: 1.000 The study explores how lattice distortion in high-entropy cation-disordered rocksalt Na-ion cathodes influences ion percolation, traditionally viewed as a topological issue linked to zero-transition-metal (0-TM) pathways. The authors demonstrate that the distortion allows kinetic accessibility to less favorable 1-TM pathways, leading to the identification of a Na-rich cathode with enhanced Na-ion transport properties. This research redefines ion percolation as a combination of topology and kinetics, emphasizing the role of lattice adaptability in optimizing ion transport in these advanced materials. Journal: *ACS Energy Letters ASAP*
- **Silicon-Assisted Direct Regeneration of Spent Lithium Iron Phosphate Cathodes via Coupled Lattice and Electronic Reactivation** — score: 1.000 The article presents a novel silicon-assisted direct regeneration method for spent lithium iron phosphate (LFP) cathodes that enhances their performance through the coupling of lattice restoration with electronic reactivation. By incorporating Si⁴⁺ into phosphorus-deficient sites, the approach reduces lithium ion migration barriers and improves charge transfer, resulting in a regenerated material (Si-RLFP) that exhibits high capacity, rapid charge capability, and excellent cycling stability. This strategy represents a promising and sustainable option for recycling LFP cathodes efficiently. Journal: *Wiley: Small Methods: Table of Contents*
- **Asymmetric Solvent Enables Adaptive Electrolyte with Orientation-Modulated Structure for Ultrafast Charging Magnesium Metal Batteries** — score: 1.000 This research introduces a novel electrolyte design using asymmetric cosolvents that enhances the performance of magnesium metal batteries under high-current conditions. By manipulating solvent orientation through electric fields, the study decreases solvation reorganization energy and facilitates faster electron transfer, leading to improved Mg²⁺ migration and stable cycling. The resulting battery configurations demonstrate impressive high-rate performances, achieving substantial capacity at elevated current densities. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Additive-Mediated Disruption of Al³⁺ Hydration Shells Stabilizes Reversible Aqueous**

Aluminum Batteries — score: 1.000 The study presents hexamethylphosphoramide (HMPA) as an effective additive to enhance the performance of aqueous aluminum-ion batteries by disrupting the strong hydration shells around Al^{3+} ions. By coordinating with Al^{3+} , HMPA decreases water activity at the electrode interface, significantly improving aluminum surface wetting and reducing corrosion, resulting in notable gains in electrochemical performance, including a specific capacity of 194.9 mAh/g and excellent cycle stability. Journal: *ACS Applied Energy Materials* ASAP

- **Characterizing Biopolymer Electrolytes in Zinc–Air Batteries: Challenges, Best Practices, and a Robust Workflow Guiding Future Research Paths** — score: 0.900 The article reviews the challenges and best practices related to the characterization of bio-based gel polymer electrolytes (GPEs) for zinc–air batteries, emphasizing how inconsistent characterization methods hinder progress in this sustainable technology. By categorizing various characterization strategies and addressing specific operational challenges, the review offers a comprehensive framework designed to enhance the development of efficient and reliable GPEs, thereby promoting advancements in sustainable energy storage solutions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Hydration Asymmetry and Solvent-Mediated Charge Balancing in Microporous Carbons: An NMR Study of Aqueous and Organic Electrosorption** — score: 0.900 This study employs in situ and MAS NMR spectroscopy to explore the effects of aqueous and organic triflate electrolytes on charge storage in commercial activated carbon supercapacitors. It reveals that hydration asymmetry and acidic in-pore environments significantly influence ion solvation and water dissociation, ultimately affecting the electrochemical performance and capacitance of the system. Journal: *Journal of the American Chemical Society* ASAP
- **A Three-Descriptor Framework for Solvation Design of Low-Temperature Electrolytes in Lithium Batteries** — score: 0.900 The article presents a novel framework for designing low-temperature electrolytes for lithium batteries, linking molecular structure to Li^{+} –solvent interactions through three key descriptors. By screening various solvents, the study identifies 2,2,2-trifluoroethylmethanesulfonate (TM) as a promising candidate that enhances solvation and reduces kinetic barriers, leading to highly efficient electrolytes capable of operating effectively at subzero temperatures. Journal: *Journal of the American Chemical Society* ASAP
- **Staggered Quadruple-Rutile Oxides as Low-Strain and High-Power Anodes for Lithium-Ion Batteries** — score: 0.900 The staggered quadruple-rutile oxide framework, specifically $\text{Li}_1\text{–Cr}_1\text{–Ti}_1\text{O}_4$, demonstrates exceptional structural integrity and electrochemical properties, enabling it to reversibly accommodate 1.15 lithium ions per formula unit, yielding a capacity of approximately 183 mAh g^{–1} with minimal volume change during lithiation. Its low activation energy for lithium diffusion and steady solid-solution reaction make it a strong candidate for high-performance anodes in lithium-ion batteries. Journal: *Journal of the American Chemical Society* ASAP
- **Atomic-Scale Intergrowth Regulation Enables Long-Cycling Zero-Strain Sodium Layered Oxide Cathodes** — score: 0.900 The study investigates P2-type layered oxide cathodes, addressing the tradeoff between capacity and structural stability common in conventional methods. By employing density functional theory and careful tuning of transition metals, the researchers developed a P2/O3-93 cathode that achieves a capacity increase to 112 mAh g^{–1}, exhibits minimal volume change, and maintains high cycling stability with 93% retention after 500 cycles, demonstrating its potential for enhanced performance in sodium-ion batteries. Journal: *ACS Energy Letters* ASAP
- **Orbital Occupancy Engineering of Co d-Band Centers via Interfacial Electron Injection Enables High-Efficiency Alkaline Zinc Batteries** — score: 0.900 The study introduces a novel approach to enhance cobalt sulfide as a cathode for alkaline zinc batteries by modulating its d-orbital configuration through interfacial electron transfer from a $\text{Ti}_3\text{C}_2\text{T}_x$ MXene substrate. This modification improves the material’s OH^{-} adsorption, conductivity, and redox kinetics, resulting in a battery that achieves a high areal capacity of 1.69 mAh cm^{–2}, excellent cycling stability, and notable energy and power densities. Journal: *ACS Applied Energy Materials* ASAP
- **Particle Size and Composition Coupling Determine the Interfacial Stability in Sulfide**

All-Solid-State Battery Composite Cathodes — score: 0.900 This study examines how the particle size of the Li₆PS₅Cl solid-state electrolyte (SSE) and the composite active material (CAM) to SSE ratio influence interfacial stability in all-solid-state batteries (ASSBs). Through simulations and characterization, it finds that a 50:50 CAM/SSE ratio optimizes ionic continuity and electronic percolation, improving electrochemical stability and capacity retention while mitigating issues associated with uneven cathode electrolyte interphase formation and microcracking. Journal: *ACS Applied Energy Materials* ASAP

- **Atomically Dispersed Manganese Electrocatalyst for Efficient Solid–Solid Conversion in Li–Se Battery** — score: 0.900 The article presents a manganese single-atom electrocatalyst (Mn-NC) for enhancing the solid-to-solid conversion efficiency in lithium-selenium (Li-Se) batteries. By utilizing a metal-organic framework and achieving atomically dispersed manganese within a nitrogen-doped carbon matrix, the catalyst addresses the sluggish reaction kinetics and high energy barriers typically associated with selenium conversion, leading to improved discharge capacities and promoting fast, reversible redox processes. This work highlights a scalable approach to optimize chalcogen-based battery systems for better energy storage capabilities. Journal: *ACS Applied Energy Materials* ASAP
- **Chemical-Delithiation-Induced Surface Reconstruction in Layered Oxide Cathodes for High-Performance All-Solid-State Batteries** — score: 0.900 The article presents a novel surface reconstruction approach using chemical delithiation to enhance the stability of Li-rich layered oxide cathodes in all-solid-state batteries. By treating the cathodes with NO₂BF₄, a cation-disordered sub-surface layer is formed, which mitigates interfacial chemical degradation and significantly improves cycling performance, achieving a reversible capacity comparable to liquid batteries and maintaining 82.3% capacity over 1500 cycles. This method may also be applicable to Ni-rich layered oxides, highlighting its potential for broader applications in high-energy-density batteries. Journal: *Journal of the American Chemical Society* ASAP
- **Electrolyte Microenvironment and Alloy Interface Synergy Enables Wide-Temperature Aqueous Aluminum-Ion Batteries From –30 to 50°C** — score: 0.800 This study presents an innovative approach to enhance the performance of aqueous aluminum-ion batteries (AAIBs) across a wide temperature range (–30 to 50°C) by utilizing a hydrated eutectic electrolyte combined with an Al-Zn alloy anode. The synergistic effects of the electrolyte’s restructured solvation environment and the alloy’s improved interface chemistry lead to enhanced stability, reduced parasitic reactions, and significant cycling durability, showcasing a promising strategy for resilient energy storage solutions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **“Electronic Sponge” Strategy Enables Polyether Electrolytes for High-Voltage and Fast-Charging Quasi-Solid-State Batteries** — score: 0.800 The article presents a novel “electronic sponge” strategy to enhance polyether electrolytes for lithium metal batteries, addressing issues of low lithium ion conductivity and stability under high voltage and rapid charging. This approach improves ionic conductivity to 1.61 mS cm^{–1} and stabilizes interphases, resulting in significant performance improvements, including over 1000 hours of stable cycling and high capacity retention in battery cells. The findings offer a new framework for developing advanced electrolytes and interphase stability in quasi-solid-state batteries. Journal: *Journal of the American Chemical Society* ASAP
- **Ternary-Ion Batteries** — score: 0.800 The article introduces a novel ternary-ion battery mechanism that harnesses three distinct ionic species, facilitating cooperative redox reactions to enhance energy density and kinetics. Focusing on an Al/eGr-PTO battery prototype, the study demonstrates effective cycling stability, minimal electrolyte uptake, and a specific energy of 84 Wh kg^{–1}, surpassing existing dual-ion battery technologies. This innovative approach aims to redefine the landscape of electrochemical energy storage by overcoming the limitations of traditional single- and dual-ion systems. Journal: *Journal of the American Chemical Society* ASAP
- **Restraining O–O Dimerization for Highly Reversible Oxygen Redox in a Sodium-Ion Layered-Oxide Cathode** — score: 0.800 This study presents a novel strategy to enhance the reversibility of oxygen redox reactions in sodium-ion layered-oxide cathodes by inhibiting O–O dimerization through the introduction of a -type conjugated ring via Sc³⁺ doping. This modification results

in significantly improved electrochemical performance, with the optimized $\text{Na}_{0.67}\text{Fe}_{0.3}\text{Sc}_{0.1}\text{Mn}_{0.6}\text{O}_2$ demonstrating a high capacity retention after multiple cycles and a pouch cell exhibiting a competitive energy density of 254.3 Wh kg^{-1} . Journal: *Journal of the American Chemical Society ASAP*

- **Intramolecular Hydrogen-Bonded Networks for Tuning Anthraquinone Redox Properties** — score: 0.800 The study focuses on designing and synthesizing intramolecular hydrogen-bonded networks to modulate the redox properties of anthraquinone derivatives, specifically through the use of imidazole, phenol, and benzimidazole moieties. Cyclic voltammetry demonstrates a notable +440 mV anodic shift in redox potential with network extension, while spectroelectrochemical techniques reveal shifts in absorption and infrared bands, indicating significant alterations in electronic structure linked to hydrogen-bond interactions, which can inform the development of finely-tuned catalysts. Journal: *Journal of the American Chemical Society ASAP*
- **Decoupling the Cation-Induced Vibrational Responses at the Graphdiyne–Water Interface via In Situ Surface-Enhanced Raman Spectroscopy** — score: 0.800 The study utilizes in situ surface-enhanced Raman spectroscopy and ab initio molecular dynamics simulations to explore how different cations interact with graphdiyne (GDY) at the water interface. The research reveals a nonmonotonic cation dependence affecting vibrational responses, indicating distinct behaviors for Li^+/K^+ compared to Na^+/Cs^+ . These findings enhance the understanding of charge-transfer kinetics and electric double layer structures at electrified carbon interfaces, highlighting the significance of ion coordination and hydration at the molecular level. Journal: *Journal of the American Chemical Society ASAP*
- **Electrolyte Lubricants for Solid-State Batteries** — score: 0.800 The article discusses the development of a novel electrolyte lubricant called ACE (aggregate cluster electrolyte) for solid-state batteries (SSBs), which enhances performance by addressing issues with interparticle voids and unstable solid–solid interfaces. ACE promotes the formation of a stable solid–liquid electrolyte interphase, improves lithium-ion conduction, and enables more efficient cycling under practical conditions, ultimately aiming to make sulfide-based SSBs safer and more effective. Journal: *Journal of the American Chemical Society ASAP*
- **Mixed-Linker Metal–Organic Frameworks Modified Separators as Li^+ -Transport Accelerators for Wide-Temperature and Long-Cycling Lithium Metal Batteries** — score: 0.800 The article discusses the development of mixed-linker metal-organic frameworks (MOFs) that enhance lithium ion transport in polypropylene separators, thereby addressing issues of interfacial instability and dendrite growth in lithium metal batteries. By incorporating tetrathiafulvalene ligands and specific linkers, these MOFs improve Li^+ migration and inhibit detrimental PF_6^- decomposition, leading to enhanced cycling stability and performance across a temperature range of -15 to 100°C , with batteries demonstrating over 500 stable cycles. This research offers a promising method for creating more efficient and durable lithium metal batteries. Journal: *Journal of the American Chemical Society ASAP*
- **Engineering a Fast Li^+ -Diffusion Secondary Phase in Sn Anodes for Low-Pressure All-Solid-State Batteries** — score: 0.800 This study enhances the utility of tin (Sn) anodes in low-pressure all-solid-state batteries (ASSBs) by incorporating bismuth (Bi), producing a diffusion-engineered Sn-Bi anode that achieves a high initial Coulombic efficiency of approximately 90% and a specific capacity of around 700 mAh g^{-1} . The developed anodes, characterized by superior lithium diffusion capabilities and stability in wider voltage ranges, demonstrate impressive rate performance and cycling stability when paired with a lithium nickel cobalt manganese oxide cathode, showcasing their potential for practical ASSB applications. Journal: *Journal of the American Chemical Society ASAP*
- **Compositional Effects on Proton and Oxygen Ion Transport in $\text{BaCo}_{1-x}\text{Fe}_x\text{O}_{3-\delta}$ (BCFZY)** — score: 0.800 The study investigates the transport characteristics of protons and oxygen ions in the triple-conducting material $\text{BaCo}_{1-x}\text{Fe}_x\text{O}_{3-\delta}$ (BCFZY), focusing on the effects of varying the Co/Fe ratio. Utilizing high-throughput density functional theory and neutron diffraction, the authors reveal that an increased Co content enhances oxygen vacancy concentration and

reduces diffusion energy barriers, while identifying key proton sites that influence transport dynamics. This research highlights the complex relationship between material composition and ionic transport properties in BCFZY. Journal: *Journal of the American Chemical Society ASAP*

- **Morphology-Diversified Sn Growth for Aqueous Acidic Batteries** — score: 0.800 The study addresses dendritic growth issues in Sn anodes for aqueous acidic batteries by introducing a porous carbon layer rich in N-defect sites, which regulates charge distribution and Sn nucleation. This innovative approach enhances the uniformity of Sn deposition, resulting in a remarkable Coulombic efficiency of 99.72% over 16,000 cycles and 83.5% capacity retention after 20,000 cycles in pouch cells, indicating significant improvements in the stability and reversibility of Sn anodes. Journal: *Journal of the American Chemical Society ASAP*
- **Bifunctional Electrolyte Additive Enables In Situ Solid Electrolyte Interphase Formation and Restrains Polyiodides Shuttling for High-Efficiency Zinc-Iodine Batteries** — score: 0.800 The article discusses the use of sodium alginate as a bifunctional electrolyte additive in aqueous zinc-iodine batteries to enhance their performance. By facilitating the formation of a stable solid electrolyte interphase at the zinc anode and reducing polyiodide shuttling at the iodine cathode, the modified batteries demonstrated a significantly improved cycling lifespan and discharge capacity, making them more viable for energy storage applications. Journal: *Energy & Fuels ASAP*
- **Shielding Surface Oxygen via Cationic Charge Anchorage To Stabilize Reversible Anionic Redox in Lithium-Rich Layered Oxides** — score: 0.800 The article discusses a strategy to enhance the stability of lithium-rich layered oxides by using large-radius cesium ions to form a protective charge-shielding layer that mitigates the reactivity of undercoordinated surface oxygen. This approach reduces the risk of premature electron extraction and gas evolution, thereby improving the overall performance and longevity of these materials in high-voltage applications. Journal: *ACS Energy Letters ASAP*
- **A Multifunctional Lithium Amide Salt for Circular Battery Recycling** — score: 0.800 The article presents a new lithium trifluoro-N-methylacetamide (LiTFMA) salt designed to enhance the stability and lifecycle of lithium-ion batteries by mitigating issues caused by moisture and hydrofluoric acid formation. This innovative approach not only rejuvenates the ionic conductivity and improves the cathode-electrolyte interphase but also enables the regeneration of retired battery electrodes, achieving over 80% capacity retention after 300 cycles, paving the way for a more sustainable circular battery economy. Journal: *ACS Energy Letters ASAP*